

Name of the Student	P. Hithaishi
Reg. No	192425196
GitHub Link	https://github.com/hithu14/Deep-learning-MLA0403

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME:Deep Learning
SLOT: A

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%
Assessment Tool Weightage: 30%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I P. Hithaishi (192425196) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name of the student:

Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation (2.5)	Critical Analysis & Technical Evaluation (2)	Future Scope, Presentation & Documentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Total						

ARTICLE REVIEW**Selected Topic: Bidirectional RNN / Bidirectional LSTM****Research Article****“Image captioning using bidirectional LSTM neural network”**

The selected article proposes a 14-layer Bidirectional LSTM (Bi-LSTM) model for automatic image caption generation. A lightweight Mobile Net CNN is used to extract visual features, while Bi-LSTM processes the sequential language information in both forward and backward directions. The model is evaluated using Flickr8k and MS COCO datasets and compared with five alternative deep-learning architectures. The study is directly relevant to RNN, Bidirectional RNN, LSTM and Encoder–Decoder/Seq2Seq concepts.

1. Article Details

Item	Details
Title	Image captioning using bidirectional LSTM neural network
Authors	Farnaz Hoseini and Anaram Yaghoobi Notash
Journal	Discover Artificial Intelligence
Year	2025
Volume	5
Article Number	80
Publication Date	29 May 2025
DOI	10.1007/s44163-025-00315-8
Technique	Bidirectional LSTM / RNN
Application	Automatic Image Captioning
Implementation	Python, TensorFlow, Google Colab
CNN Feature Extractor	MobileNet pretrained using ImageNet

The article was published on 29 May 2025 in Discover Artificial Intelligence.

2. Research Problem

The research addresses the problem of automatically generating meaningful textual descriptions for images.

An image contains visual information such as:

- Objects
- People
- Actions
- Background
- Spatial relationships
- Scene information

Converting this visual information into a meaningful natural-language sentence is difficult because the system must understand both visual features and language sequences.

Traditional manual image captioning is:

- Time-consuming
- Expensive
- Subjective
- Difficult to scale to large image collections

The authors therefore investigate an automatic image-captioning system that combines CNN-based image feature extraction with a Bidirectional LSTM language model.

Limitations of Existing Approaches

The article discusses several problems with earlier approaches:

1. Conventional RNNs have difficulty learning long-term dependencies.
2. Unidirectional LSTMs process information only in one direction.
3. Image-captioning systems may have difficulty understanding complex visual contexts.
4. Large CNN architectures can require substantial computational resources.
5. Generating coherent captions requires both visual understanding and sequential language modeling.
6. Manual image annotation is not practical for very large image collections.

3. Objectives of the Study

The main objectives are:

1. To develop an automatic image-captioning system using Bidirectional LSTM.
2. To combine MobileNet-based visual feature extraction with Bi-LSTM sequential processing.
3. To improve contextual understanding of captions by processing sequences in both directions.
4. To reduce computational requirements through the use of MobileNet.
5. To compare the proposed Bi-LSTM model with other deep-learning architectures.
6. To generate captions that are more relevant to the visual content of an image.

The model is intended to provide a balance between caption quality, computational efficiency and contextual understanding.

4. Dataset Description

The research uses two major image-captioning datasets:

A. Flickr8k

The Flickr8k dataset contains approximately 8,092 images.

Each image is associated with five human-written captions.

The images contain common real-world scenes involving:

- People
- Animals
- Objects
- Outdoor activities
- Sports
- Everyday events

The dataset is divided into training, development and testing image lists. The article specifically refers to files such as:

- Flickr_8k. trainImages
- Flickr_8k. devImages
- Flickr_8k. testImages
- Flickr8k.token

B. MS COCO

The article also uses the Microsoft COCO dataset.

MS COCO contains more than 330,000 images, with more than 200,000 labeled images and multiple human-written captions for images. It contains diverse real-world scenes and objects.

The paper describes the combined data used in the experiments as exceeding 1.04 GB and containing more than 8,000 images in the specific dataset setup discussed.

Type of Data

The research uses multimodal data:

Image data + Text/caption data

Therefore, the model has to learn a relationship between:

Visual information → Language sequence

5. Proposed Methodology

The proposed methodology has three major stages:

Stage 1 – Image Feature Extraction

The input image is processed using MobileNet.

MobileNet is a lightweight CNN that extracts important visual features such as:

- Shapes
- Objects
- Textures
- Spatial information

The authors use pretrained ImageNet knowledge to obtain useful visual representations.

Stage 2 – Caption Processing

The words in captions are converted into numerical representations using an Embedding layer.

For example:

"dog" → numerical vector

"running" → numerical vector

"water" → numerical vector

The resulting vectors can be processed by the recurrent network.

Stage 3 – Bi-LSTM Processing

The visual features and textual information are provided to the Bi-LSTM architecture.

The Bi-LSTM has:

- Forward LSTM
- Backward LSTM

The forward network processes the sequence from left to right.

The backward network processes it from right to left.

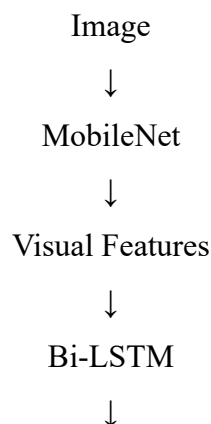
Their outputs are combined to obtain richer contextual information.

Stage 4 – Caption Generation

The combined Bi-LSTM representation is passed through Dense layers and a SoftMax output.

The model predicts the next word in the caption.

For example:



" A "
↓
" dog "
↓
" is "
↓
" running "
↓
" on "
↓
" water "

6. Deep Learning Technique Used

Bidirectional LSTM

The primary technique is Bidirectional Long Short-Term Memory (Bi-LSTM).

A normal LSTM processes information in one direction:

Word 1 → Word 2 → Word 3 → Word 4

A Bidirectional LSTM processes information in both directions:

Forward:

Word 1 → Word 2 → Word 3 → Word 4

Backward:

Word 4 → Word 3 → Word 2 → Word 1

The two hidden states are combined:

$$h_t = [\rightarrow h_t ; \leftarrow h_t]$$

This allows the model to use information from both directions when learning contextual relationships.

Why LSTM?

A conventional RNN can suffer from the vanishing-gradient problem when learning long sequences.

LSTM addresses this using memory cells and gates:

Forget Gate

Determines which previous information should be discarded.

Input Gate

Determines what new information should be stored.

Output Gate

Determines what information should be passed to the next stage.

Therefore, LSTM can preserve useful information over longer sequences.

Why Bidirectional LSTM?

A normal LSTM uses only previous information.

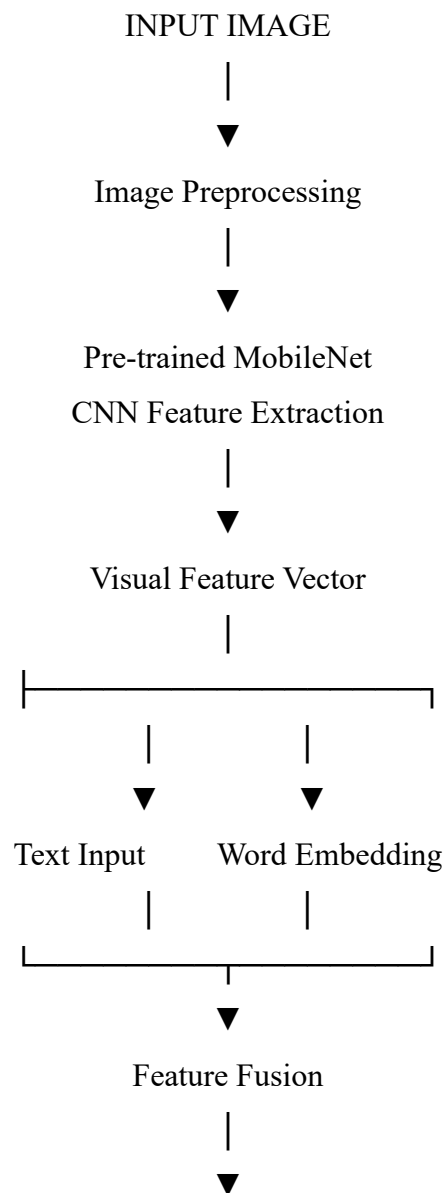
Bi-LSTM uses information from both:

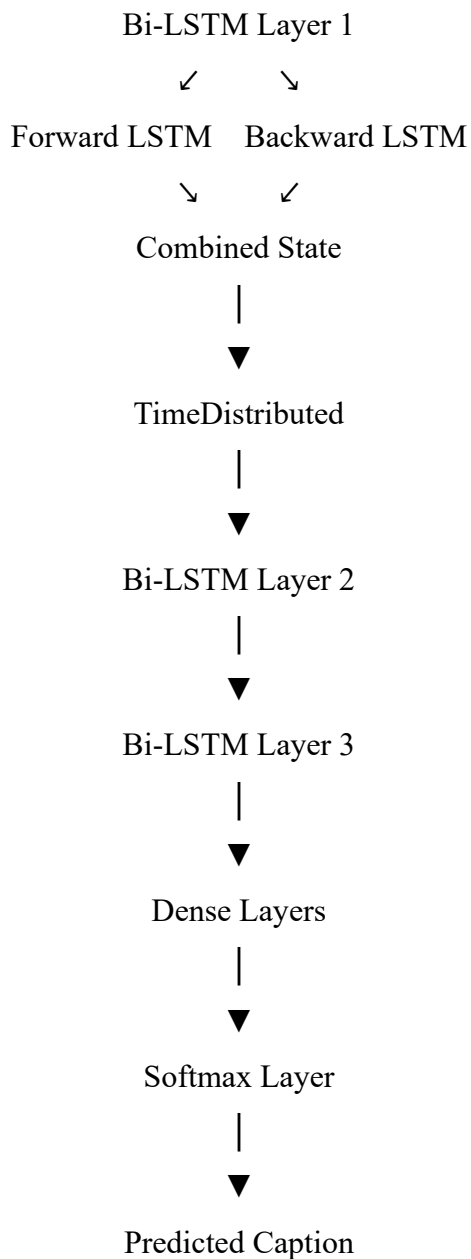
Past context + Future context

This can improve the understanding of relationships between words in a caption. The article specifically motivates Bi-LSTM because it can capture contextual information in both directions and handle long-term dependencies.

7. Model Architecture

The simplified architecture is:





The paper describes a 14-layer architecture consisting of input layers, embedding, dropout, Bi-LSTM layers, Dense layers, TimeDistributed processing, an Add/Merge layer and a final Softmax layer. The network has more than 9 million trainable parameters.

Major Components

Component	Function
MobileNet	Extracts image features
Embedding	Converts words into numerical vectors
Bi-LSTM	Learns forward and backward sequence dependencies
Dropout	Reduces overfitting
Time Distributed	Maintains sequential processing
Add/Merge	Combines visual and textual information
Dense	Learns higher-level relationships

SoftMax	Predicts the next word/output
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The model uses three Bi-LSTM layers and dropout values including 0.2 and 0.5. RMSprop is reported as the optimizer.

8. Performance Evaluation

The authors evaluate the proposed model using:

- Accuracy
- Precision
- Recall
- F1-score
- Loss function

The formulas given in the article include:

Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Accuracy

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

F1-score

$$\text{F1} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

Major Performance Result

The proposed Bi-LSTM model achieved an overall reported accuracy of 75.90%.

The authors state that this was approximately 3% higher than the fourth comparison model and that the proposed model also achieved a reduced loss value of approximately 2.56.

The authors report that the proposed architecture achieved the highest values among their five comparison models under their experimental conditions.

Comparison

The article reports the following approximate results for the comparison architectures:

Model	Reported Accuracy
Model 1	92%
Model 2	Not clearly reported
Model 3	Lower than Model 1
Model 4	About 44%
Proposed 14-layer Bi-LSTM	75.90%
Model 5	About 7%

Critical note: The paper's reporting of these comparison values is not completely consistent with its statement that the proposed model has the highest accuracy, because it also mentions a 92% result for one comparison model. This is an important issue to consider when critically reviewing the article.

9. Results and Discussion

The main findings of the research are:

1. Bi-LSTM Can Capture Bidirectional Context

The model can process textual information from both directions, helping it understand contextual relationships.

2. MobileNet Reduces Computational Complexity

Instead of using a very large CNN, MobileNet is used for feature extraction. This reduces computational requirements.

3. Layer Arrangement Is Important

The researchers experimented with different numbers and arrangements of LSTM/Bi-LSTM layers.

An important finding was:

Simply increasing the number of layers does not necessarily improve performance.

The 21-layer comparison model performed poorly, while the proposed 14-layer arrangement performed better.

4. Dropout Helps Reduce Overfitting

Dropout was inserted at different locations in the architecture to reduce overfitting.

5. Proposed Model Achieved 75.90% Accuracy

The proposed model achieved the reported 75.90% accuracy despite the hardware limitations encountered during experimentation.

10. Advantages of the Proposed Model

1. Captures Bidirectional Context

Bi-LSTM processes information in both directions, allowing better contextual understanding than a standard unidirectional RNN.

2. Handles Long-Term Dependencies

LSTM memory cells allow the model to retain important information across a sequence.

3. Combines Vision and Language

The model combines MobileNet visual features with textual information, making it suitable for multimodal image-captioning tasks.

4. Lightweight CNN Backbone

MobileNet requires fewer computational resources than many larger CNN architectures. The authors specifically use it to reduce the feature-extraction burden.

5. Overfitting Control

Multiple dropout layers are used to reduce overfitting.

6. Useful for Accessibility

Automatic image descriptions can assist visually impaired users by converting visual information into text.

11. Limitations of the Study

1. Limited GPU Resources

The authors identify insufficient GPU resources as their main limitation.

Google Colab's available GPU memory caused long execution times and prevented the researchers from performing more experiments. Approximately **7.43 GB GPU memory** was used during one stage of experimentation.

2. Moderate Accuracy

The reported accuracy of **75.90%** is promising but still leaves considerable room for improvement for complex image-captioning applications.

3. Dataset Limitations

Flickr8k is relatively small compared with modern large-scale image-captioning datasets. The authors also acknowledge that Flickr8k does not represent specialized domains.

4. Computational Complexity

Although MobileNet reduces CNN computation, the proposed network contains multiple Bi-LSTM layers and more than 9 million trainable parameters.

5. Limited Generalization

A model trained mainly on general-purpose image-captioning datasets may not perform equally well on specialized images such as:

- Medical images
- Satellite images
- Technical diagrams
- Industrial images

6. Lack of Modern Attention/Transformer Architecture

Modern image-captioning systems commonly use attention mechanisms and Transformer-based architectures. The proposed approach relies primarily on LSTM/Bi-LSTM.

7. Evaluation Metrics Could Be Improved

For image captioning, standard metrics such as BLEU, METEOR, ROUGE-L and CIDEr would provide more meaningful information about the linguistic quality of generated captions. The article instead emphasizes accuracy, precision, recall, F1-score and loss.

12. Critical Analysis

Is Bi-LSTM Appropriate?

Yes, but with some limitations.

Bi-LSTM is appropriate because captions are sequential text. The model needs to understand relationships between words, and LSTM is designed for sequential data.

However, for modern image captioning, a Transformer-based decoder with attention may be more appropriate because Transformers can model long-range relationships more effectively and can directly attend to different image regions.

Is the Dataset Adequate?

The Flickr8k dataset is useful for academic experimentation, but it is relatively small.

MS COCO is much larger and more diverse, so using larger portions of COCO would improve the reliability of the evaluation.

A stronger experiment would use:

Flickr8k → Initial experiment

MS COCO → Main evaluation

Flickr30k/Visual Genome → External validation

Are the Evaluation Metrics Appropriate?

This is one of the major weaknesses.

Accuracy, precision and recall can be useful for classification, but image captioning is a language-generation problem.

Therefore, the following metrics would be more suitable:

- BLEU
- METEOR
- ROUGE-L
- CIDEr

- SPICE

The paper itself discusses BLEU, METEOR and CIDEr in its architectural/ablation discussion, but the main experimental results emphasize accuracy, precision, recall, F1 and loss.

Therefore, I would recommend adding standard caption-generation metrics.

Are the Results Convincing?

The results are partially convincing.

Positive aspects:

- The model is compared with alternative architectures.
- The authors conduct architectural comparisons.
- Hardware limitations are explicitly discussed.
- The model demonstrates reasonable captioning performance.

However, there are concerns:

1. The reported comparison values are not completely consistent.
2. The use of classification-style accuracy for caption generation is questionable.
3. The dataset description is somewhat unclear regarding exactly how Flickr8k and MS COCO are jointly used.
4. More independent evaluation would strengthen the conclusions.
5. Modern Transformer-based baselines would make the comparison more meaningful.

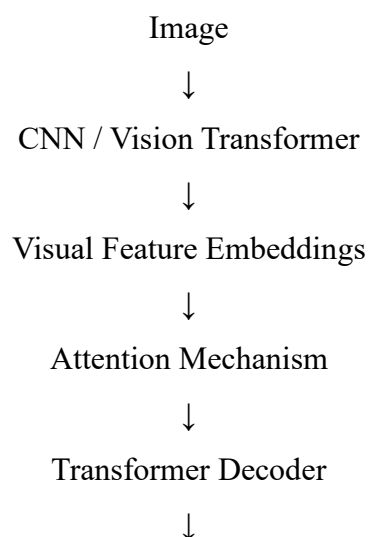
Therefore, the article provides a useful experimental study, but its claims should be interpreted cautiously.

Could Another Unit IV Technique Provide Better Performance?

Yes.

An Encoder–Decoder architecture with Attention and Transformer could potentially outperform the Bi-LSTM approach.

A modern architecture could be:



Generated Caption

Attention allows the decoder to focus on different parts of the image when generating different words.

For example:

"dog" → focus on dog region

"running" → focus on dog/action

"water" → focus on water/background

This could produce more precise captions than a Bi-LSTM alone.

13. Future Scope

The research can be improved in several directions.

1. Transformer-Based Architecture

Replace or combine Bi-LSTM with Transformer-based models.

2. Attention Mechanism

Add visual attention so the model can identify the relevant image region for each generated word.

3. Larger Dataset

Train and evaluate using larger datasets such as:

- MS COCO
- Flickr30k
- Visual Genome

4. Better Evaluation Metrics

Include:

- BLEU-1 to BLEU-4
- METEOR
- CIDEr
- ROUGE-L
- SPICE

5. Better Hardware

The authors recommend using more powerful servers so that more training iterations and experiments can be performed.

6. Multimodal Transformers

Combine image and language representations using a multimodal Transformer.

7. Transfer Learning

Use newer pretrained vision-language models rather than relying only on MobileNet/ImageNet features.

8. Beam Search

Instead of selecting the most probable word at every step, beam search could be used to explore multiple possible caption sequences.

9. Human Evaluation

Human evaluators could assess:

- Caption relevance
- Fluency
- Grammar
- Completeness
- Semantic similarity

10. Real-Time Deployment

The lightweight MobileNet component makes the approach potentially useful for mobile and embedded applications, but further optimization would be necessary for real-time deployment. The authors themselves suggest attention mechanisms, Transformers, multimodal architectures, stronger hardware and newer CNN architectures as future directions.

14. Conclusion

The reviewed article, “Image captioning using bidirectional LSTM neural network,” presents a 14-layer Bi-LSTM architecture for automatic image caption generation.

The system combines MobileNet for visual feature extraction with Bi-LSTM for sequential language processing. The Bi-LSTM processes language information in both forward and backward directions, enabling the model to capture richer contextual information.

The model uses Flickr8k and MS COCO image-captioning datasets and incorporates embedding, dropout, TimeDistributed, Add/Merge, Bi-LSTM and Dense/Softmax layers. The proposed system reports an accuracy of 75.90% and a loss value of approximately 2.56 under the experimental conditions.

The major strengths are its bidirectional contextual modeling, combination of visual and textual information, lightweight MobileNet feature extraction and use of dropout for regularization.

However, the research has limitations involving GPU resources, dataset scale, moderate performance and evaluation methodology. In particular, future studies should use standard

image-captioning metrics such as BLEU, METEOR, CIDEr and SPICE and compare the model with modern attention and Transformer-based architectures.

Overall, the article is a good example of applying Bidirectional LSTM to a multimodal sequence-generation problem, and it is highly relevant to the Unit IV topics of RNN, Bidirectional RNN, LSTM and Encoder–Decoder/Seq2Seq.

15. References

Main Research Article

Hoseini, F., & Notash, A. Y. (2025). **Image captioning using bidirectional LSTM neural network**. *Discover Artificial Intelligence*, 5, 80. <https://doi.org/10.1007/s44163-025-00315-8>.

[Full Article – Springer Nature](#)

Additional References

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6. Howard, A. G., Zhu, M., Chen, B., et al. (2017). **MobileNets: Efficient convolutional neural networks for mobile vision applications**. arXiv:1704.04861.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis